

PART-2

Step1: Update proc.h

added code:

```
int userflag;
```

Step 2: Update syscall.h

added code:

```
#define SYS_setflag 26  
#define SYS_getflag 27
```

Step 3: Update syscall.c

added code:

```
extern int sys_setflag(void);  
extern int sys_getflag(void);
```

```
[SYS_setflag] sys_setflag,  
[SYS_getflag] sys_getflag,
```

Step 4: Update sysproc.c

added code:

```
int  
sys_setflag(void)  
{  
    int val;  
    if(argint(0, &val) < 0)  
        return -1;  
  
    myproc()->userflag = val;  
    return 0;  
}
```

```
int  
sys_getflag(void)  
{  
    return myproc()->userflag;  
}
```

Step 5: Update user.h

added code:

```
int setflag(int);  
int getflag(void);
```

Step 6: Update usys.S

added code:

```
SYSCALL(setflag)
```

```
SYSCALL(getflag)
```

Step 7: Make testflags.c

code:

```
#include "types.h"
```

```
#include "stat.h"
```

```
#include "user.h"
```

```
int
```

```
main(void)
```

```
{
```

```
    // First set
```

```
    setflag(12340920);
```

```
    printf(1, "After first set: %d\n", getflag());
```

```
    // Second set
```

```
    setflag(12904321);
```

```
    printf(1, "After second set: %d\n", getflag());
```

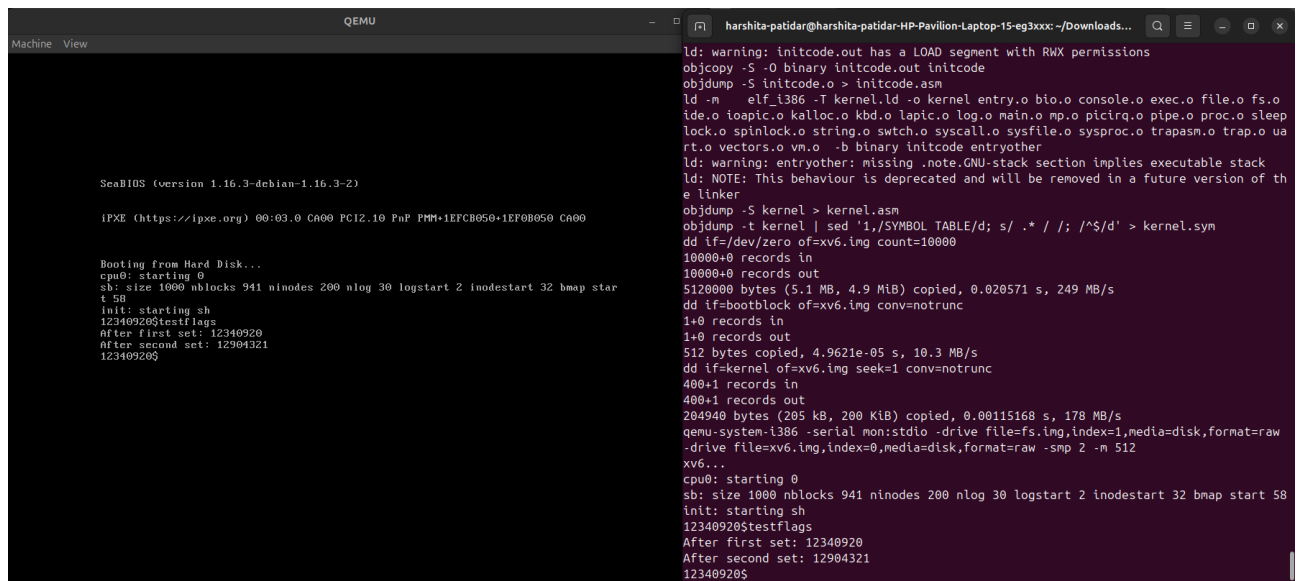
```
    exit();
```

```
}
```

FINAL STEP: MAKEFILE

added code:

_testflags



```
QEMU
Machine View
harshita-patidar@harshita-patidar-HP-Pavilion-Laptop-15-eg3xxx: ~/Downloads...

ld: warning: initcode.out has a LOAD segment with RWX permissions
objcopy -S -O binary initcode.out initcode
objdump -S initcode.o > initcode.asm
ld -m elf_i386 -T kernel.ld -o kernel entry.o bio.o console.o exec.o file.o fs.o
ide.o ioapic.o kalloc.o kbd.o lapic.o log.o main.o mp.o picirq.o pipe.o proc.o sleep
lock.o spinlock.o string.o switch.o syscall.o sysfile.o sysproc.o trapasm.o trap.o ua
rt.o vectors.o vm.o -b binary initcode entryother
ld: warning: entryother: missing .note.GNU-stack section implies executable stack
ld: NOTE: This behaviour is deprecated and will be removed in a future version of th
e linker
objdump -S kernel > kernel.asm
objdump -t kernel | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > kernel.sym
dd if=/dev/zero of=xv6.img count=10000
10000+0 records in
10000+0 records out
5120000 bytes (5.1 MB, 4.9 MiB) copied, 0.020571 s, 249 MB/s
dd if=bootblock of=xv6.img conv=notrunc
1+0 records in
1+0 records out
512 bytes copied, 4.9621e-05 s, 10.3 MB/s
dd if=kernel of=xv6.img seek=1 conv=notrunc
400+1 records in
400+1 records out
204940 bytes (205 kB, 200 KiB) copied, 0.00115168 s, 178 MB/s
qemu-system-i386 -serial mon:stdio -drive file=fs.img,index=1,media=disk,format=raw
-drive file=xv6.img,index=0,media=disk,format=raw -smp 2 -m 512
xv6...
cpu0: starting 0
sb: size 1000 nblocks 941 ninodes 200 nlog 30 logstart 2 inodestart 32 bmap start 58
init: starting sh
12340920$testflags
After first set: 12340920
After second set: 12904321
12340920$
```